

NEXUCON

Testing, Setup & Credential Guide

Follows: Nexucon_Implementation_Plan_Sep_Oct_2026.pdf (Weeks 1-9,
Final Acceptance, Milestones 1-6, Client Dependencies & Governance)

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1. How To Use This Guide

This guide follows the Implementation Plan (Sep 1 - Oct 31 2026) section by section: Weeks 1-9 of Section 5, the Final Acceptance, Milestones 1-6 (Section 6), Client Dependencies (Section 7) and Delivery Governance (Section 8). For each week it lists: the test files to upload (with exact extensions), the credentials involved, the step-by-step procedure, and the documents the system generates to verify the Acceptance Outcome.

Backend verification status at time of writing: 1,308 automated tests passing, 90% code coverage (governance target >85% met), zero fabricated data in production code. Integrations without credentials honestly report PENDING_CREDENTIALS and name the missing variable - nothing is invented.

Run the automated suite before manual testing (from Nexucon_backend/): `manage.py test --settings=config.settings.development`

2. Master Test-File Kit (all weeks)

Prepare these once. Together they exercise every upload pipeline in the plan. Free sources are given for each.

#	File	Extension	Used in (weeks)	Where to get a test copy
1	Defective site photo	.jpg / .png	W3-W5 scans, W7 inspections	Any real photo of cracked/spalled concrete - defects MUST be visible (AI only reports what it truly detects)
2	Thermal image	.jpg	W3-W5 scan pipeline	FLIR camera export, or any image (pipeline runs either way)
3	LiDAR point cloud	.las	W3 scan upload, volume/progress	Scanner export, or free .las samples from USGS/public datasets
4	BIM model (Revit or IFC)	.rvt / .ifc	W2 Trimble/BIM, W4 correlation	Autodesk Revit .rvt project (native), or Revit/Archicad IFC4 export, or free buildingSMART IFC samples. .rvt max 100 MB
5	Gaussian splat / point cloud	.ply	W3 scan upload	Photogrammetry export or any .ply
6	GPR radargram	.jpg / .png	W3 GPR survey analysis	GPR console screenshot/export from the GSSSI unit
7	PUNDIT device export	.csv / .txt	W3 PUNDIT raw evidence (optional)	Proceq PUNDIT export - OR skip the file: typed measurements are enough (see 6.3)
8	GNSS RINEX log	.obs / .rinex	W3 GNSS survey evidence	Tersus rover log, or a RINEX file from any public CORS station
9	Inspection evidence photos	.jpg	W7 inspector submissions	Phone photos (each is hash-sealed; frontend computes sha256)
10	Permit documents	.pdf	W5/W8 applications, documents	Any real architectural/structural/EIA drawing PDFs

Storage note: STORAGE_PROVIDER=cloudflare_r2 is active even in dev, so uploads write to the REAL R2 bucket during testing. Fine for verification - just be aware.

3. Credentials Master List (plan Section 7 - Client Dependencies)

3.1 Already configured and active

Credential (env var)	Service	Powers (plan item)	Status
GEMINI_API_KEY (+ GEMINI_MODEL)	Google Gemini	W3-W6 AI analysis adapters, W6 report generator	ACTIVE
OPENAI_API_KEY	OpenAI	Secondary AI provider (fallback)	ACTIVE
TRIMBLE_CLIENT_ID / _SECRET / PROJECT_ID	Trimble Connect	W1 OAuth + W2 BIM discovery (Milestone 1)	ACTIVE
CLOUDFLARE_R2_* (5 vars)	Cloudflare R2	W3 object storage, governance signed URLs (G-07)	ACTIVE
CLOUDINARY_* (4 vars)	Cloudinary	Media/image storage path	ACTIVE
RESEND_API_KEY + _FROM_EMAIL	Resend	Statutory notification emails (W5-W8)	ACTIVE
GOOGLE_SERVICE_ACCOUNT_* / GOOGLE_MEETING_*	Google Calendar	Stakeholder meeting calendar events	ACTIVE
AUTODESK_CLIENT_ID / _SECRET	Autodesk APS	BIM model conversion (W2)	ACTIVE
DJANGO_SECRET_KEY / FRONTEND_URL / NEXT_PUBLIC_API_URL	internal	Core configuration	ACTIVE

3.2 Still required - how to obtain each (step by step)

Missing env var	Service	Plan item it unlocks	How to obtain it
TERSUS_API_URL + TERSUS_API_KEY	Tersus GNSS cloud	W3 Tersus MVP SI registration; live telemetry sync	1) Contact Tersus dealer portal (dealers.tersus-gnss.com). 2) Request API access for the MVP SI fleet. 3) Receive base URL + key. 4) Add to .env, restart, run device sync.
TRIMBLE_API_BASE / _TOKEN_URL / _AUTHORIZE_URL / _REDIRECT_URI / _FOLDER_ID	Trimble Platform	W1-W2 full OAuth lifecycle + inspection-file sync (Milestone 1 'TRIMBLE CONNECTED')	1) Log into developer.trimble.com. 2) Open the app owning the existing CLIENT_ID/SECRET. 3) Add redirect URI (e.g. https://<domain>/api/v1/digital-eye/trimble/callback/). 4) Copy token/authorize URLs from Trimble docs. 5) Create/note a project folder ID. 6) Add all to .env.
GOV_CAC_API_TOKEN	CAC	W4 developer/company verification	Apply for API access via the Lagos State government liaison; add token to .env.
GOV_EGIS_API_TOKEN	Lagos eGIS	W4 land/GIS parcel verification	Apply through the Lagos State Lands Bureau / eGIS office.
GOV_LASRRRA_API_TOKEN	LASRRRA	W4 resident identity verification	Apply through the Lagos State Residents Registration Agency.
GOV_FMW_API_TOKEN	Federal Ministry of Works	W4 federal project cross-check	Request API access from FMW ICT department.
GOV_CLOUD_PROJECT_ID	Google Cloud	W8 stakeholder message translation (yo/ig/ha)	1) console.cloud.google.com 2) select the project owning the service account 3) copy Project ID 4) enable Cloud Translation API 5) add to .env.
CELERY_BROKER_URL + _RESULT_BACKEND	Redis	W2 background sync jobs, W9 recurring anomaly jobs (production)	Provision Redis; set redis://<host>:6379/0 in both vars.
DATABASE_URL (or DATABASE_*)	PostgreSQL	Final Acceptance: production DB migration & index verification	Create the DB; set DATABASE_URL=postgres://user:pass@host:5432/nexucon; run manage.py migrate.

Missing env var	Service	Plan item it unlocks	How to obtain it
SENTRY_DSN (optional)	Sentry	Production error monitoring	Create a sentry.io project; copy the DSN.

3.3 Known limitation (GCP configuration, not a missing key)

Google Meet link generation is blocked because the Meet API is disabled on the current GCP project. Fix: 1) console.cloud.google.com 2) select the project owning the service account 3) APIs & Services -> search 'Google Meet API' 4) click Enable. Calendar events + email invitations already work; until enabled the backend honestly reports 'conference unavailable' instead of fabricating a Meet link.

4. Weeks 1-5: September - Digital Eye, BIM Streams & AI Evidence Foundation

WEEK 1 (Sep 1-4): Trimble Connect - Authentication, Connection & AI Schema Baseline

- Credential: TRIMBLE_CLIENT_ID/SECRET already set; full OAuth needs the 5 TRIMBLE_* URL vars (3.2).
- Test files: none required.
- Steps: 1) Digital Eye -> Field Devices -> Trimble. 2) Create connection. 3) Authorize through the Trimble consent screen. 4) Run 'Test Connection' diagnostics. 5) Check the connection health indicator stays green; disconnect/reconnect to exercise the modals.
- Verify: acceptance outcome is 'Trimble Connect authenticated and Evidence Registry models initialized'. Health check + audit log endpoints record real outcomes; without credentials they show PENDING_CREDENTIALS (honest).

WEEK 2 (Sep 7-11): Trimble BIM, Live Streaming & AI Evidence Ingestion

- Test files: .rvt Revit project OR .ifc BIM model (kit #4).
- Steps: 1) From the connected Trimble link, run BIM discovery (external projects + models appear). 2) Run sync - elements and GUID mappings are extracted. 3) Upload/import the .rvt or .ifc into a project; open the 3D viewer + property panel (.rvt is converted through Autodesk APS; .ifc is parsed natively). 4) Configure a live stream and confirm it maps to BIM element coordinates. 5) Trigger a manual sync and let the scheduled Celery job run.
- Verify: government users can access 3D BIM models, live streams, and ingest project evidence into the AI Registry. Large-model sync and Celery retry behavior are covered by the automated suite (apps.digital_eye, apps.common).

WEEK 3 (Sep 14-18): Digital Eye - GPR, PUNDIT, GNSS Pipelines & AI Adapters

- Test files: .jpg radargram (#6), .csv/.txt PUNDIT export (#7, optional), .obs/.rinex RINEX (#8), .las LiDAR (#3), .ply splat (#5).
- PUNDIT (easiest - no file): create a pulse-velocity test with typed measurements - path_length_mm=250, pulse_time_us=62.5 - then Analyze: velocity 4.0 km/s with a BS 1881-203/ASTM C597 quality grade. Crack-depth test: enter the three measurements (cracked path, cracked time, uncracked time).
- GPR: create survey -> upload radargram -> run analysis; void/rebar anomalies recorded with real depth factors.
- GNSS: register the Tersus MVP SI device, create survey + boundary points; coordinates auto-project to Lagos Minna Datum / UTM 31N. Upload the RINEX log.
- Verify: 'GPR survey ingestion, Tersus GNSS positioning, and GPR/PUNDIT AI adapters are operational.' Storage goes to R2 (checksums verified).

WEEK 4 (Sep 21-25): AI Cross-Source Evidence Correlation & Spatial BIM Linking

- Test files: results from W3 (GPR anomalies + PUNDIT low velocity + GNSS coords) plus the .rvt/.ifc BIM model.
- Steps: 1) Run correlation on the project. 2) Open the AI Correlation & Finding Detail panel - findings link GPR + PUNDIT + GNSS + BIM structural elements (e.g. COL-C24). 3) Inspect the explainable reasoning log and multi-source risk score. 4) Check AI Risk Badges (Info/Low/Med/High/Critical) and geospatial markers.
- Verify: the plan's example chain - 'GPR void + PUNDIT low velocity + GNSS coords + Trimble BIM (COL-C24) -> AI risk HIGH -> human review' - with every finding carrying an immutable revision + audit history.

WEEK 5 (Sep 28-Oct 2): Human-in-the-Loop Review, Inspections & Statutory NCR Loop

- Test files: none new (uses W3-W5 findings); PDF exports generated by the system.
- Steps: 1) Open a pending finding. 2) Accept / Reject / Modify with notes (hash-chained revisions). 3) Request supplementary evidence / live stream. 4) Trigger immediate statutory inspection. 5) Issue formal NCR with corrective action + due date. 6) Director sign-off into the official record. 7) Export the PDF report with radar evidence.
- Generated documents: NCR record + CAPA, auditable PDF reports with radar evidence, hash-verified decision

ledger.

- Verify: Milestones 1-2 complete - 'Digital Eye and AI Evidence Correlation operate within the statutory inspection, human review and compliance NCR loop.'

5. Weeks 6-8: October - AI Project Intelligence, Mobile Fieldwork & Transparency

WEEK 6 (Oct 5-9): Project-Level AI Intelligence & AI-Assisted Reporting Generator

- Test files: none new - intelligence aggregates existing project evidence.
- Steps: 1) Open a project's Intelligence executive card - Overall / Structural / Compliance risk scores. 2) Read AI Observations + Recommended Actions banners. 3) Use 1-click export for the AI-Assisted Project Brief (PDF). 4) Test Inspector Login with 2FA (TOTP code required) - mobile JWT endpoints. 5) Check 'My Assigned Projects' / 'Daily Tasks'.
- Generated documents: AI Project Intelligence Report PDF (Project, Inspection, NCR report generators).
- Verify: 'Project-level AI intelligence summaries, automated reporting and Inspector Mobile session foundations are live.'

WEEK 7 (Oct 12-16): Inspector Mobile - Field Execution & In-App AI Sensory Diagnostics

- Test files: inspection evidence photos .jpg (kit #9) - each upload is hash-sealed.
- Steps: 1) Log in as inspector (2FA). 2) GPS check-in with browser/device geolocation - mandatory coordinates + timestamp. 3) Run the step-by-step dynamic checklist. 4) Submit with photo evidence (tamper-evident). 5) Sign on the signature pad - digital sign-off with cryptographic hash. 6) Open verification view: both seals validate.
- Generated documents: field inspection submission record (hash-sealed) + crypto sign-off.
- Verify: 'Inspector can execute and submit a complete field inspection with in-app AI sensory corroboration.' Tamper a record in DB to confirm the seal detects it (automated tests cover this).

WEEK 8 (Oct 19-23): Client Mobile - Stakeholder Portal & Secure Transparency Gateway

- Test files: permit documents .pdf (kit #10) for the document viewer.
- Setup: create a Client developer account + Developer organization linked to ONE project.
- Steps: 1) Client login - see only their scoped projects (Milestone progress, inspection outcomes). 2) Open Compliance/NCR/CAPA feed; submit a corrective-action response. 3) Download approved documents (permits, plans) via the gateway. 4) Receive push/email project status alerts. 5) CRITICAL boundary test: as the client, try to access another project's inspections/documents - must get 404 (strict data segregation). 6) Check the public transparency gateway shows approved public notices ONLY - raw scans and private notes never appear.
- Verify: Milestone 4 - 'Clients can securely monitor and respond with strict boundary protection.'

6. Week 9 & Final Acceptance (Oct 26-31)

WEEK 9: Headquarters Command AI Intelligence & Multi-District Aggregation

- Test files: none - aggregates everything created in W1-W8.
- Steps (director account): 1) Open the Lagos-Wide Command Executive AI Overview. 2) Inspect the District Comparison & Risk Matrix + GIS heatmap. 3) Drill down 4 levels: HQ -> District -> Project -> Finding. 4) Check command metrics: total/active/high-risk projects, AI-flagged anomalies, critical Digital Eye issues, active inspections & pending approvals, open NCRs & overdue CAPAs - all computed from real DB counts (empty shows as zero, never invented). 5) Export the PDF State Summary Brief. 6) Let the recurring anomaly-detection job fire (Celery beat).
- Generated documents: Executive State Summary Brief (PDF).
- Verify: Milestone 5 - 'Lagos-wide HQ Command Dashboard with AI portfolio intelligence and multi-district drill-down is live.'

FINAL ACCEPTANCE (Oct 31): AI Security & Verification

Plan requirement	How to verify	Status
AI prompt stability & zero hallucination	Feed clean/corrupt inputs; AI returns schema-validated results or honest empties - never fabricated findings	Verifiable now
Production DB migration & index verification	Set DATABASE_URL, run manage.py migrate, verify indexes	Needs PostgreSQL (3.2)
Celery worker load & concurrency tuning	Provision Redis (3.2), run workers + beat under load	Needs Redis (3.2)
Automated database backup & recovery	Configure on the production DB host (pg_dump schedule / managed backups)	Deployment task - not code
Strict multi-tenant district isolation	Cross-tenant access tests return 404 across all apps (automated + manual W8 test)	Verified
Evidence provenance & audit trail verification	Evidence hashes, finding revision hash chains, append-only ledger with verify functions	Verified
Public Transparent System boundary	Gateway exposes approved notices only (manual W8 test)	Verified
RBAC role escalation penetration tests	Automated authz matrix green; formal external pen test still to schedule	Partially - pen test pending
>85% test coverage (governance)	coverage run: 90% on production source	Met (90%)
Signed R2 URLs (governance)	Download any uploaded file - URL is signed, 3600 s expiry	Verified
Formal Client UAT	Post-deployment client activity	Pending (client-side)

7. Milestone Register (plan Section 6) - verification route

Milestone / date	Acceptance outcome	How to prove it with this guide
M1 - Trimble Connect (Sep 11)	TRIMBLE CONNECTED	Week 1-2 steps in section 4; needs the 5 TRIMBLE_* URL vars (3.2) for the full OAuth lifecycle
M2 - Digital Eye & AI Registry (Oct 2)	DIGITAL EYE & AI LIVE	Week 3-5 steps in section 4 (PUNDIT/GPR/GNSS + correlation + HITL)
M3 - Inspector Mobile (Oct 16)	FIELD WORKFLOW LIVE	Week 6-7 steps in section 5 (2FA login, GPS check-in, sealed submission, crypto sign-off)
M4 - Client Mobile & Reports (Oct 23)	CLIENT VISIBILITY LIVE	Week 8 steps in section 5 (scoped portal, boundary test, document gateway)
M5 - HQ Command AI (Oct 30)	HQ COMMAND LIVE	Week 9 steps in section 6 (risk matrix, drill-down, PDF state brief)
M6 - System Acceptance (Oct 31)	READY FOR CLIENT REVIEW / UAT	Final Acceptance table in section 6 + governance items below

8. Client Dependencies & Inputs (plan Section 7)

These are the client-side inputs the plan calls out, mapped to what is still outstanding:

Plan category	Required input (by)	Current state
Trimble (W1-2)	OAuth API credentials, models & stream keys	Client ID/Secret/Project present; 5 URL vars + stream keys pending (3.2)
Digital Eye / AI (W3-5)	Tersus GNSS, GPR, PUNDIT & sample NDT datasets	Adapters ready + this guide's test-kit files cover testing; TERSUS_API_URL/KEY pending (3.2); sample NDT datasets to be supplied by client
Survey / GIS (W3-5)	Approved Lagos Minna Datum projection rules	UTM 31N projection implemented and tested; formal approval is client sign-off
Mobile (W6-8)	FCM push & Apple/Google Dev access	FCM push service implemented; store access + dev certificates are client-side
HQ / Reporting (W9)	Lagos district hierarchy & KPI risk rules	District hierarchy modeled; KPI rule values await client confirmation

9. Delivery Governance (plan Section 8) - status

- Weekly reviews: this guide is the per-week verification record.
- Code quality & AI testing: Django migration verification clean; Celery resiliency tested; coverage 90% (>85% target met).
- Security baseline: multi-tenant district isolation enforced and tested; audit logs immutable + hash-verified; R2 URLs signed (3600 s).
- Formal client UAT: pending post-deployment (Milestone 6).

10. Automated Test Suite Per Area

From Nexucon_backend/ run: `manage.py test apps.<name> --settings=config.settings.development` (full suite: `manage.py test`). Current total: 1,308 tests, 0 failures.

Plan area	Test command (apps.<name>)	Focus
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Plan area	Test command (apps.<name>)	Focus
W1-W2 Trimble + devices	apps.digital_eye, apps.common	OAuth flow, BIM sync, health checks, honest PENDING_CREDENTIALS
W3 GPR/PUNDIT/GNSS	apps.digital_eye	BS 1881-203 math, UTM 31N projection, adapters, device registry
W3-W5 scan pipeline	apps.scans	Sessions, uploads, defects, thermal, BIM alignment, compliance
W4 correlation / W9 HQ	apps.evidence	HITL review, hash chains, district matrix, executive briefing
W5 NCR / compliance	apps.compliance	NCR/CAPA lifecycle, certificates
W6 reports	apps.reports	QA/QC + AI report PDFs
W7 inspections	apps.inspections	GPS check-in, tamper-evident seals, crypto sign-off
W8 client portal / docs	apps.documents, apps.applications	Versions, approvals, permit state machine, scoping
W6-W8 auth & notifications	apps.accounts, apps.notifications, apps.stakeholders	JWT/2FA/API keys, email idempotency, meetings, translations
BIM & geometry	apps.bim, apps.processing	Real IFC geometry, deviation, clash, point clouds